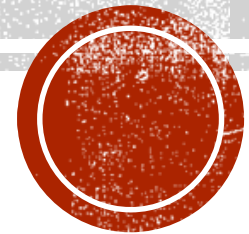


INTRODUCTION TO RECONFIGURABLE COMPUTING

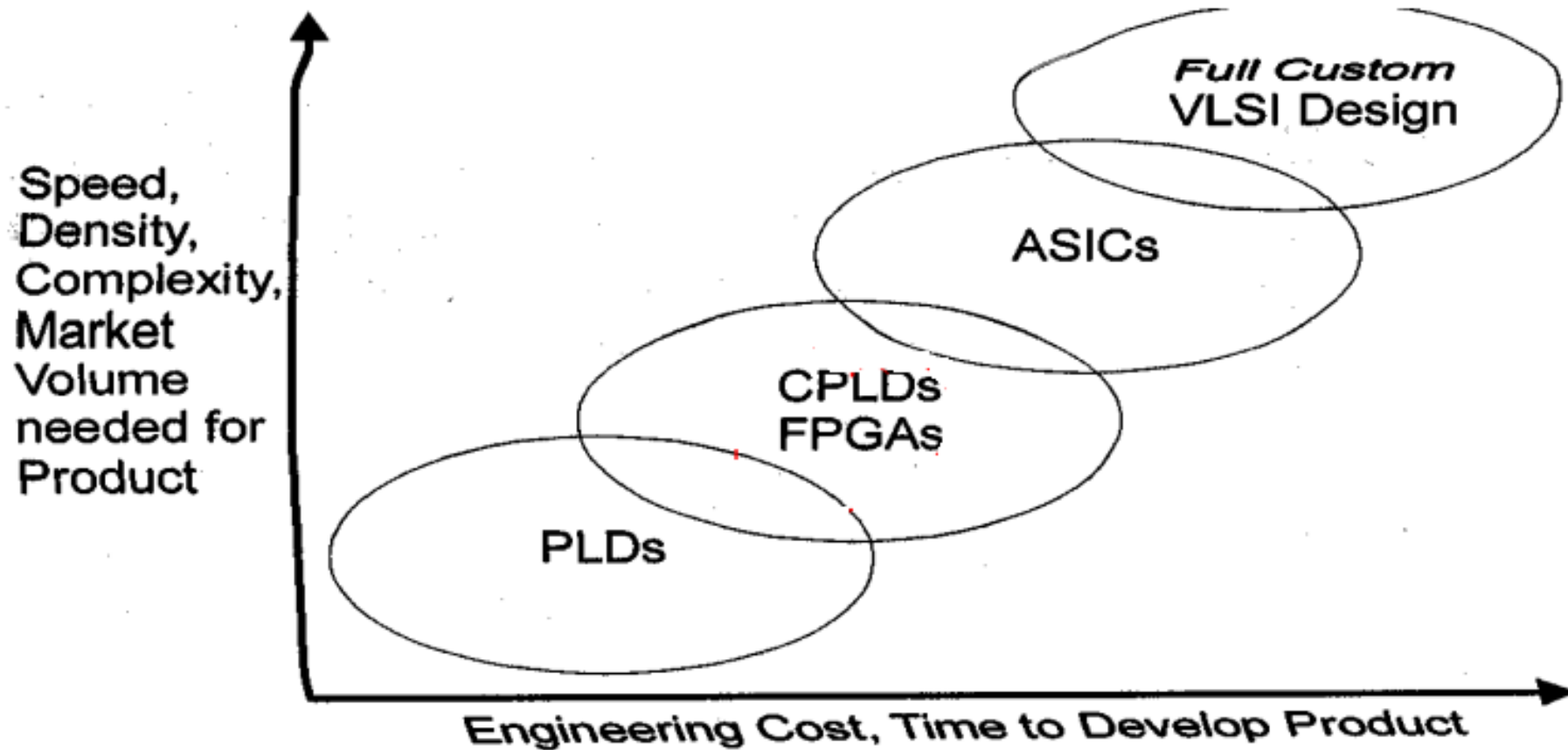
CS G553: Reconfigurable computing

Dr. Meetha V Shenoy

Lect-3



DIGITAL LOGIC

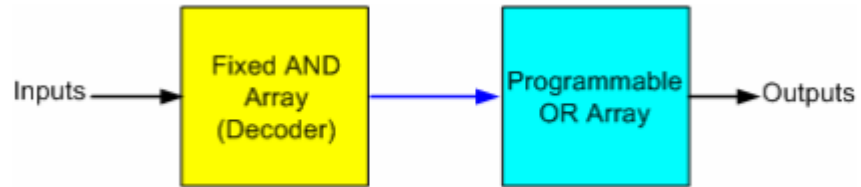


PROGRAMMABLE LOGIC DEVICES (PLD)

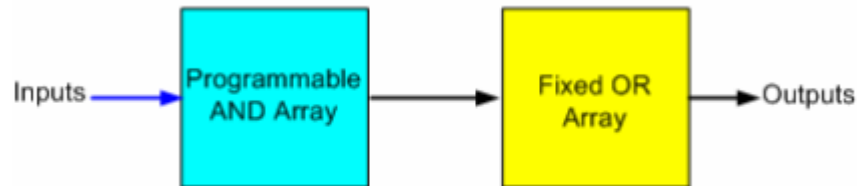
- ROM
- Programmable Logic Array (PLA)
- Programmable Array Logic (PAL)



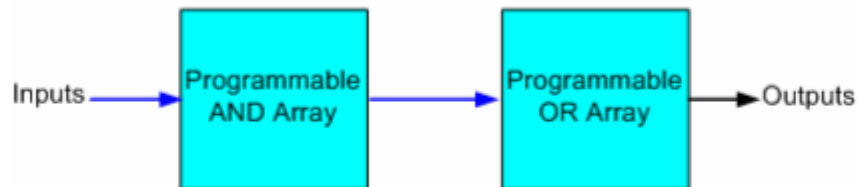
FUNDAMENTAL PLD TYPES



(a) Programmable Read Only Memory (PROM)



(b) Programmable Array Logic (PAL) Device



(c) Programmable Logic Array (PLA) Device

→ Programmable Connections

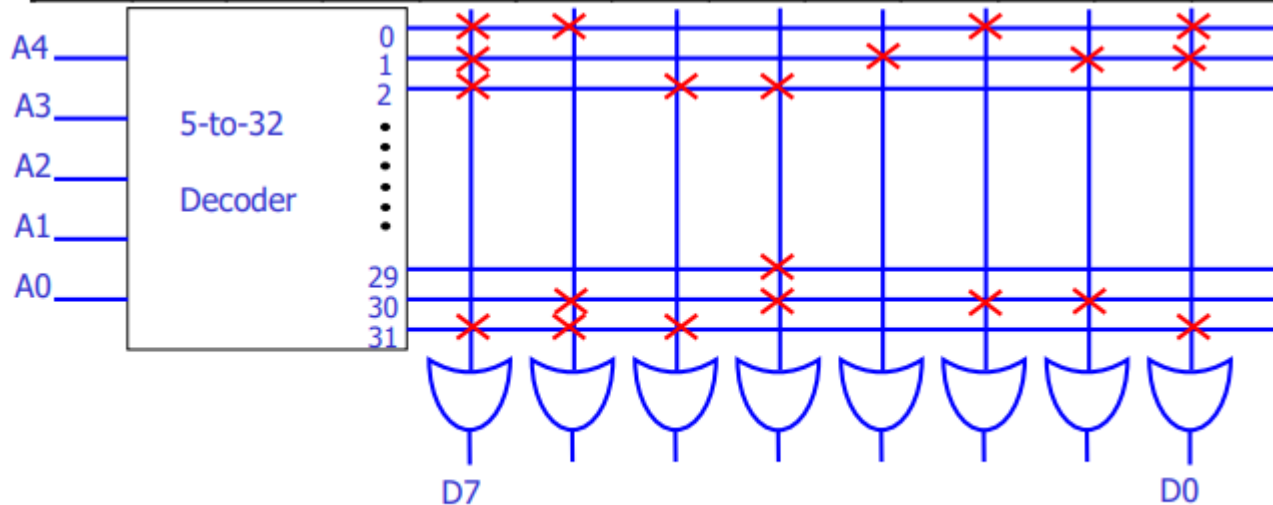
→ Normal Connections



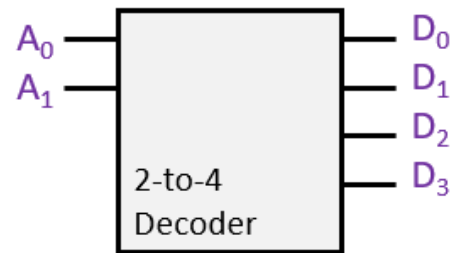
32X8 PROM

A4	A3	A2	A1	A0	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	0	1	1	0	0	0	1	0	1
0	0	0	0	1	1	0	0	0	1	0	1	1
0	0	0	1	0	1	0	1	1	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...
1	1	1	0	1	0	0	0	1	0	0	0	0
1	1	1	1	0	0	1	0	1	0	1	1	0
1	1	1	1	1	1	1	1	0	0	0	0	1

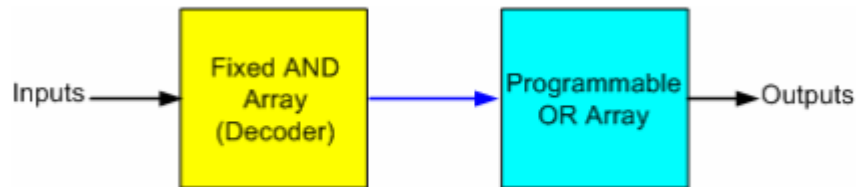
X indicates a connection



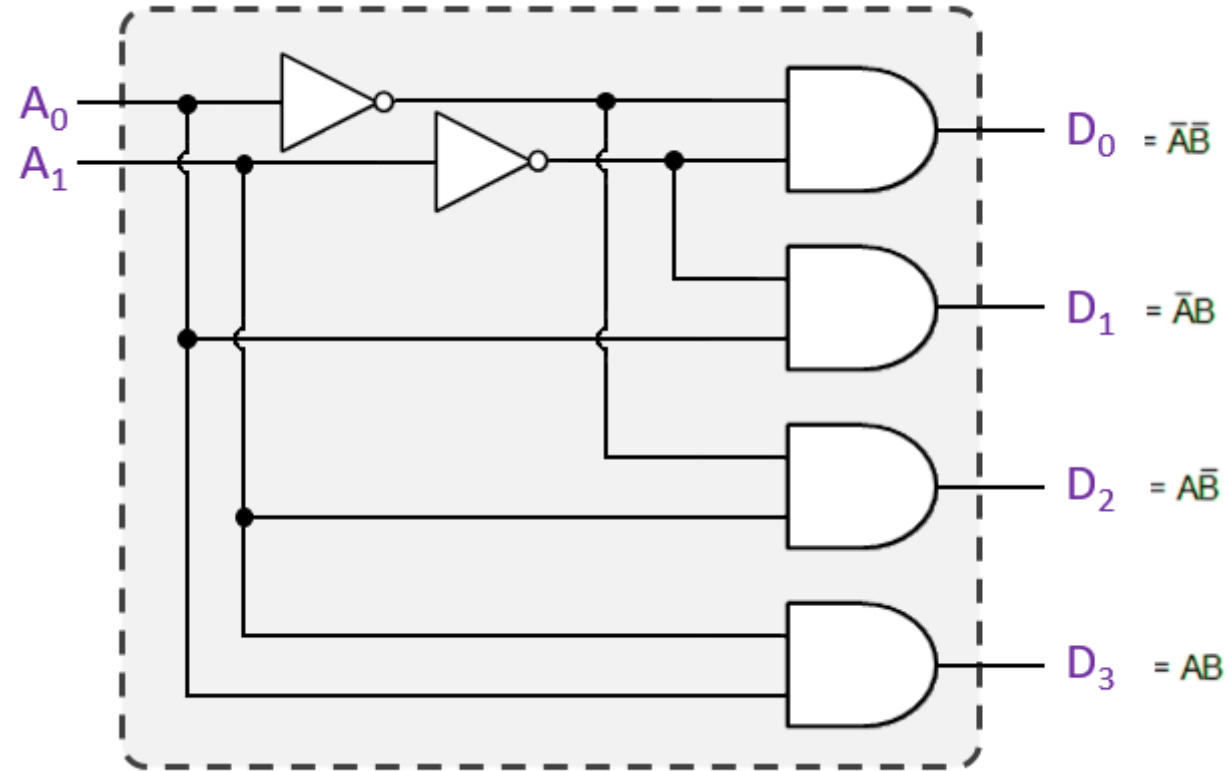
PROM AS A PLD



INPUT A_1A_0	OUTPUT $D_3D_2D_1D_0$
00	0001
01	0010
10	0100
11	1000



(a) Programmable Read Only Memory (PROM)



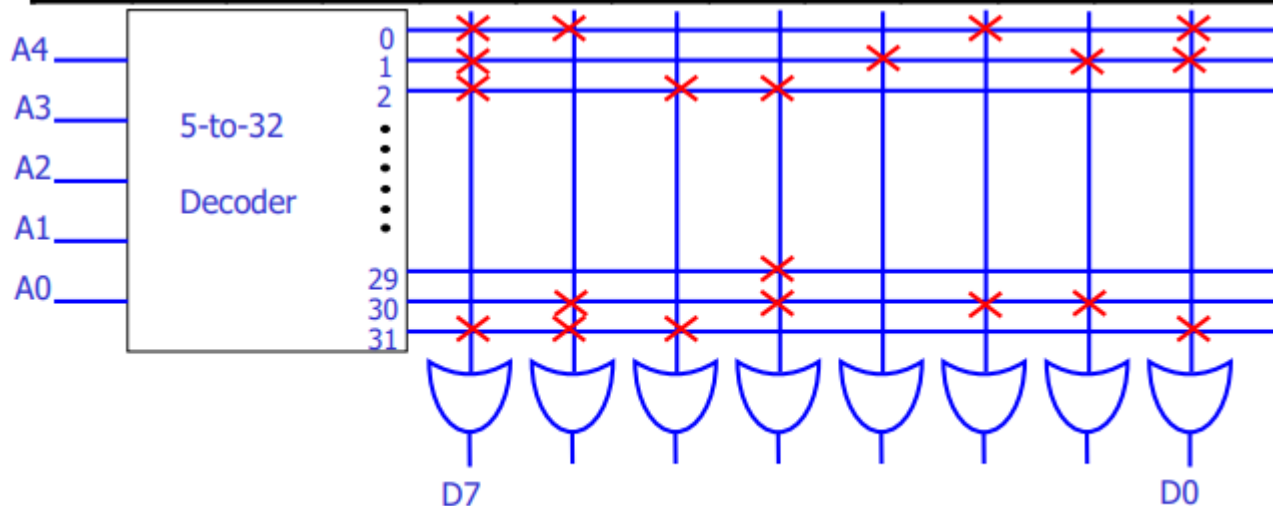
→ Programmable Connections

→ Normal Connections

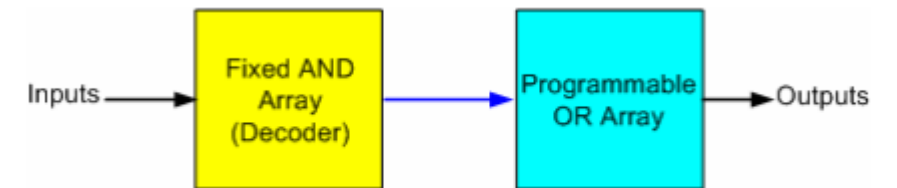


32X8 PROM

A4	A3	A2	A1	A0	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	0	1	1	0	0	0	1	0	1
0	0	0	0	1	1	0	0	0	1	0	1	1
0	0	0	1	0	1	0	1	1	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...
1	1	1	0	1	0	0	0	1	0	0	0	0
1	1	1	1	0	0	1	0	1	0	1	1	0
1	1	1	1	1	1	1	1	0	0	0	0	1



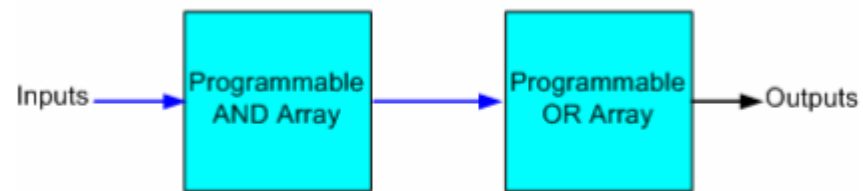
X indicates a connection



(a) Programmable Read Only Memory (PROM)



PROGRAMMABLE LOGIC ARRAY



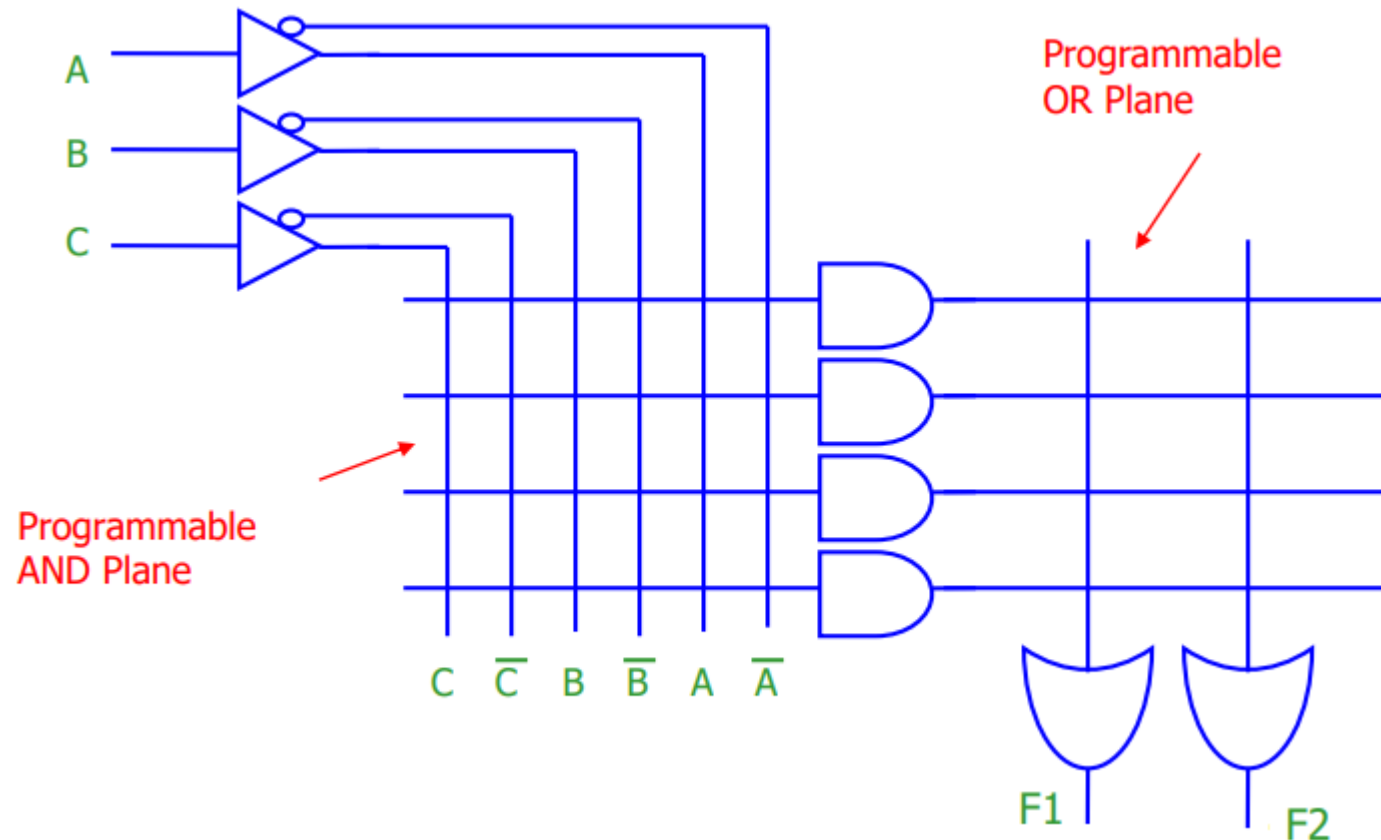
(c) Programmable Logic Array (PLA) Device

→ Programmable Connections

→ Normal Connections



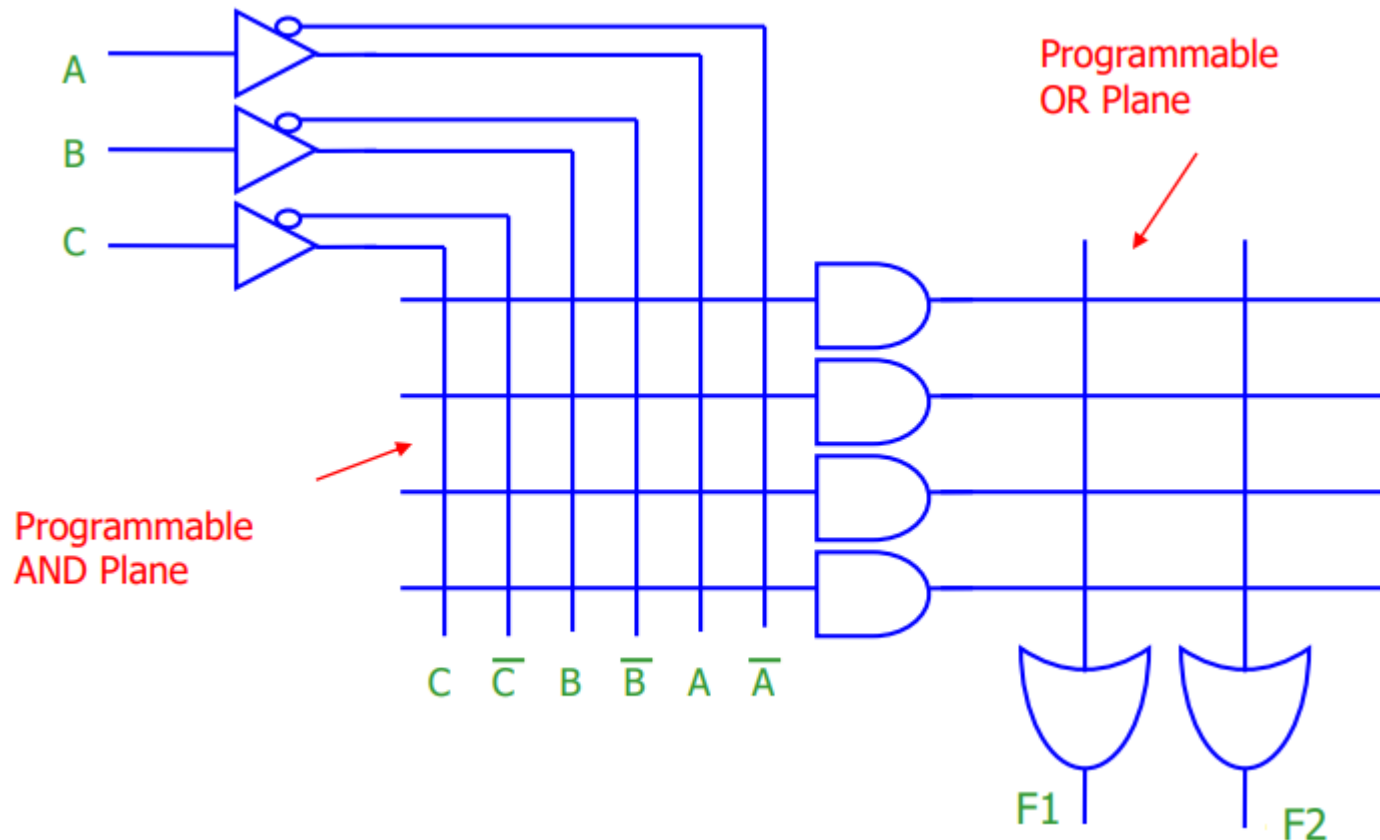
PLA



SIZE OF PLA: 3X4X2
3 INPUTS
4 PRODUCT TERMS
2 OUTPUTS



OBTAIN TRUE AND COMPLEMENTARY OUTPUT



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0



IMPLEMENT USING PLA

$$V2 = \Sigma m(1,2,3,4)$$

$$V1 = \Sigma m(2,6,7)$$

$$V0 = \Sigma m(4,6,7)$$

$$V2 = xy'z' + x'z + x'y$$

$$V1 = yz' + xy$$

$$V0 = xz' + xy$$



IMPLEMENT USING PLA

$$V2 = \Sigma m(1,2,3,4)$$

$$V1 = \Sigma m(2,6,7)$$

$$V0 = \Sigma m(4,6,7)$$

$$V2 = xy'z' + x'z + x'yz'$$

$$V1 = x'yz' + xy$$

$$V0 = xy'z' + xy$$

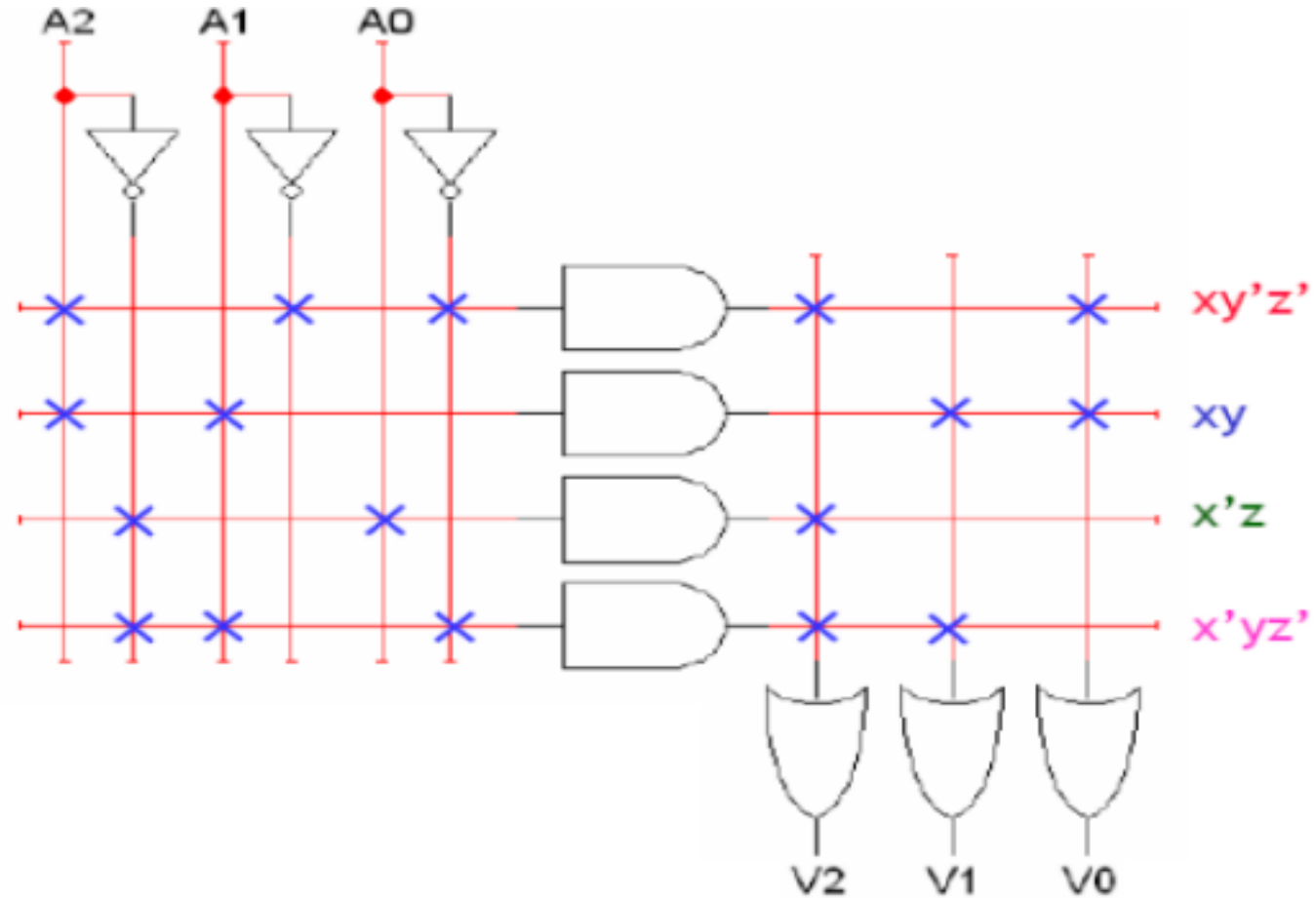


$$V2 = \Sigma m(1,2,3,4) = xy'z' + x'z + x'yz'$$

$$V1 = \Sigma m(2,6,7) = x'yz' + xy$$

$$V0 = \Sigma m(4,6,7) = xy'z' + xy$$

Size= 3x4x3



IMPLEMENT USING PLA

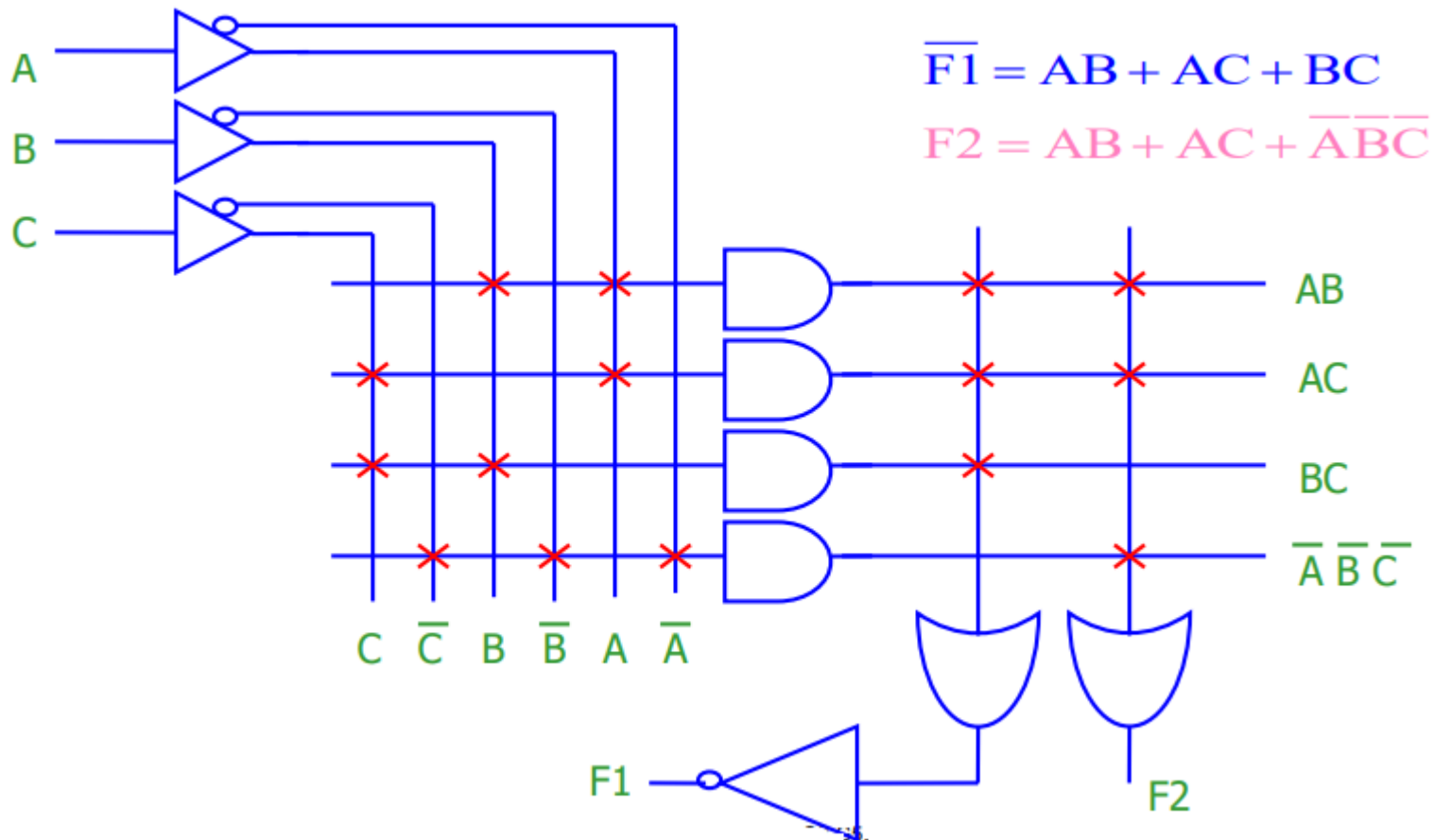
$$F1(A,B,C) = \sum m(0,1,2,4)$$

$$F2(A,B,C) = \sum m(0,5,6,7)$$

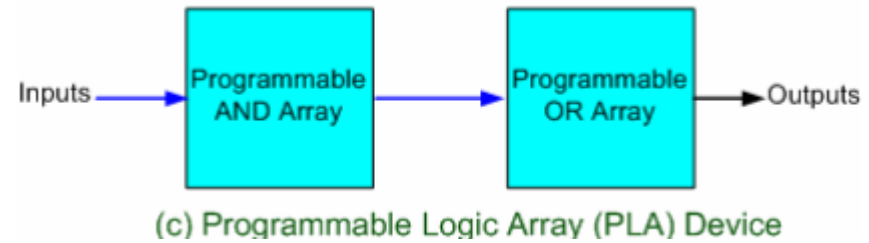
- Size of PLA ??



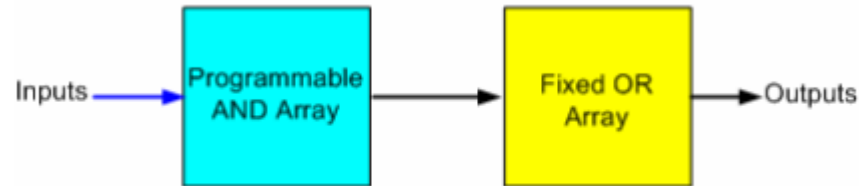
IMPLEMENT USING PLA



PLA

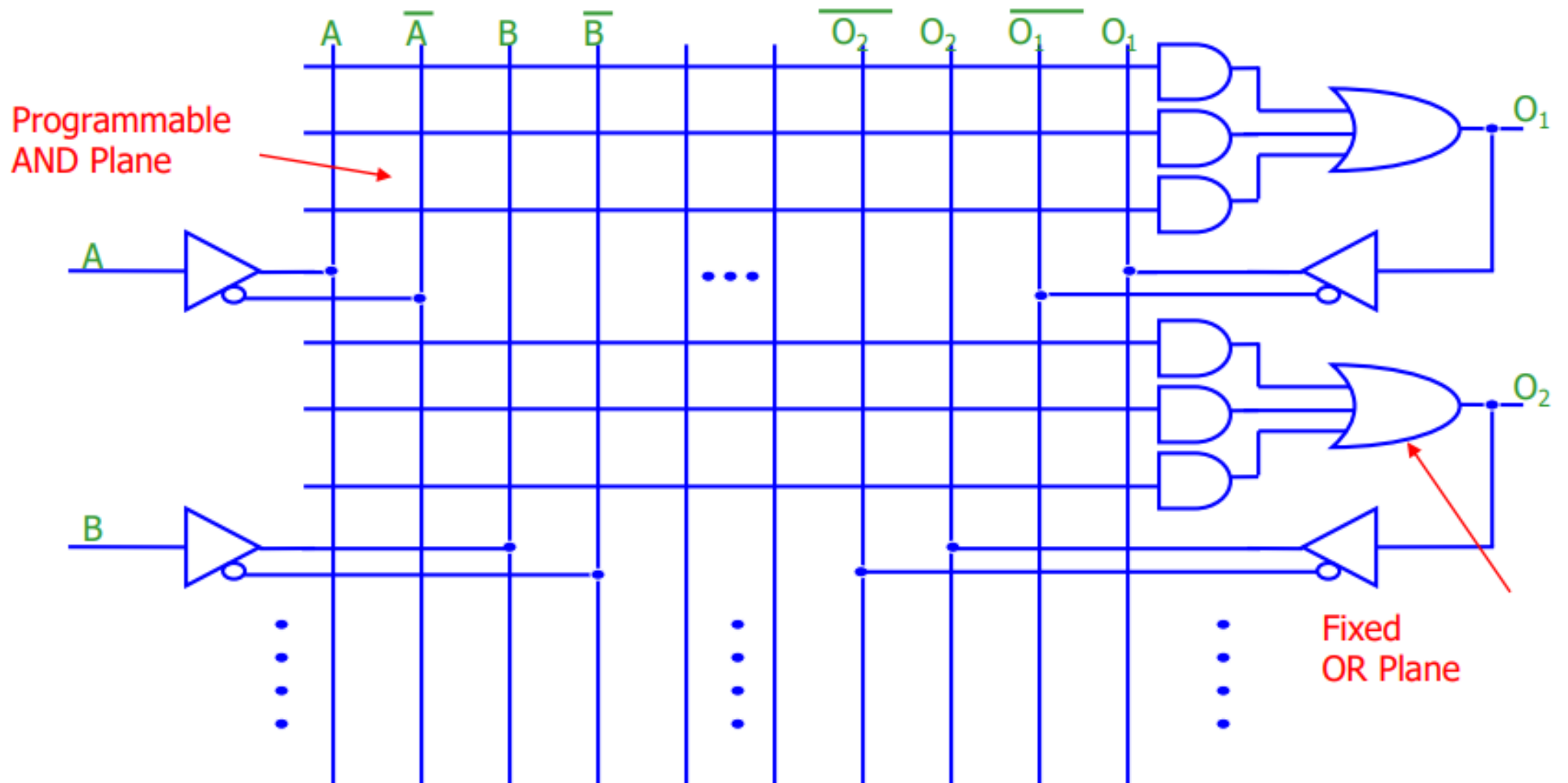


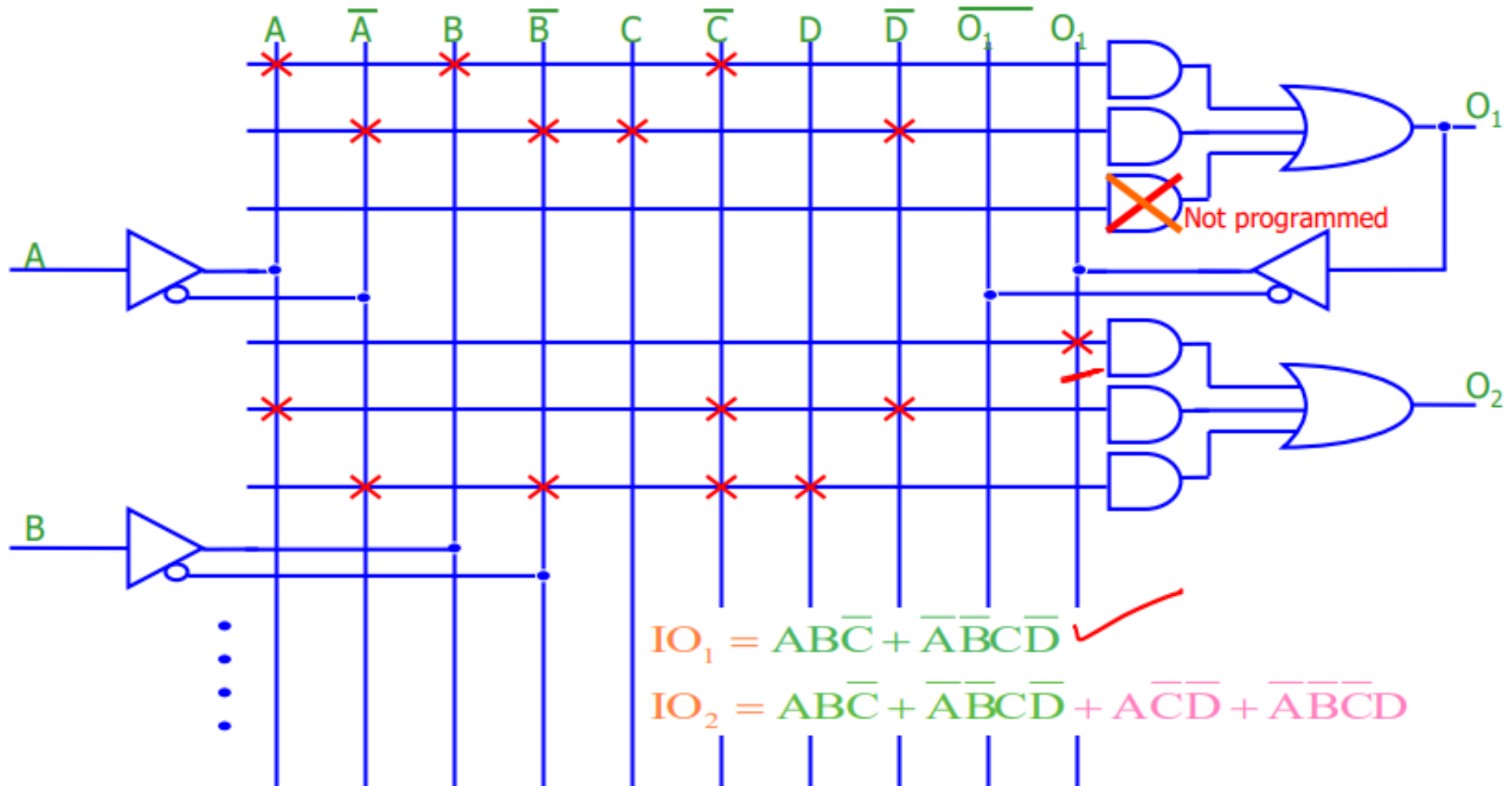
PAL



(b) Programmable Array Logic (PAL) Device







PAL

$$W(A,B,C,D) = \sum m(2,12,13)$$

$$X(A,B,C,D) = \sum m(7,8,9,10,11,12,13,14,15)$$

$$Y(A,B,C,D) = \sum m(0,2,3,4,5,6,7,8,10,11,15)$$

$$Z(A,B,C,D) = \sum m(1,2,8,12,13)$$

$$W = ABC' + A'B'CD'$$

$$X = A + BCD$$

$$Y = A'B + CD + B'D'$$

$$\begin{aligned} Z &= ABC' + A'B'CD' + AC'D' + A'B'C'D \\ &= W + AC'D' + A'B'C'D \end{aligned}$$



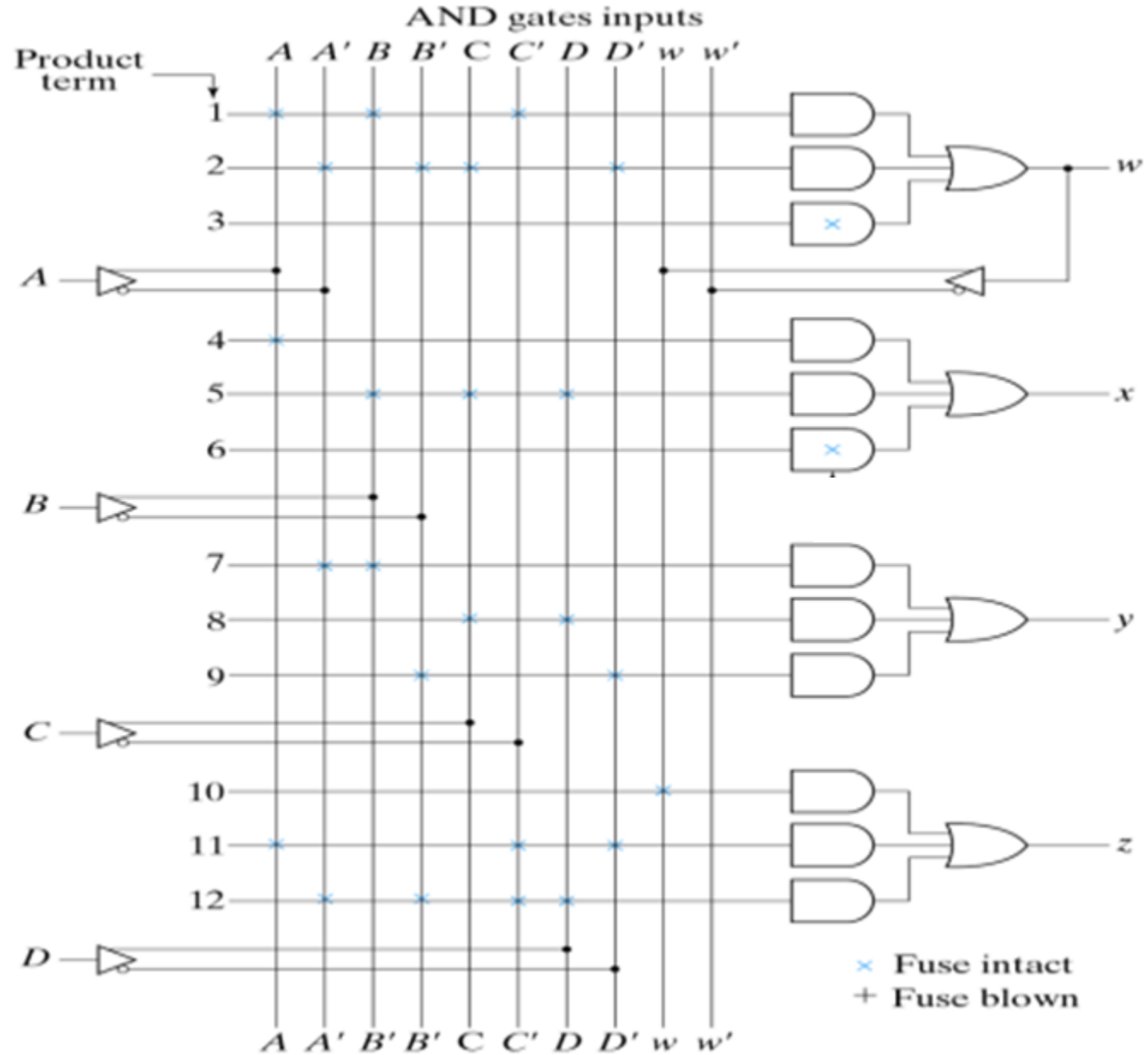
$$W = ABC' + A'B'CD'$$

$$X = A + BCD$$

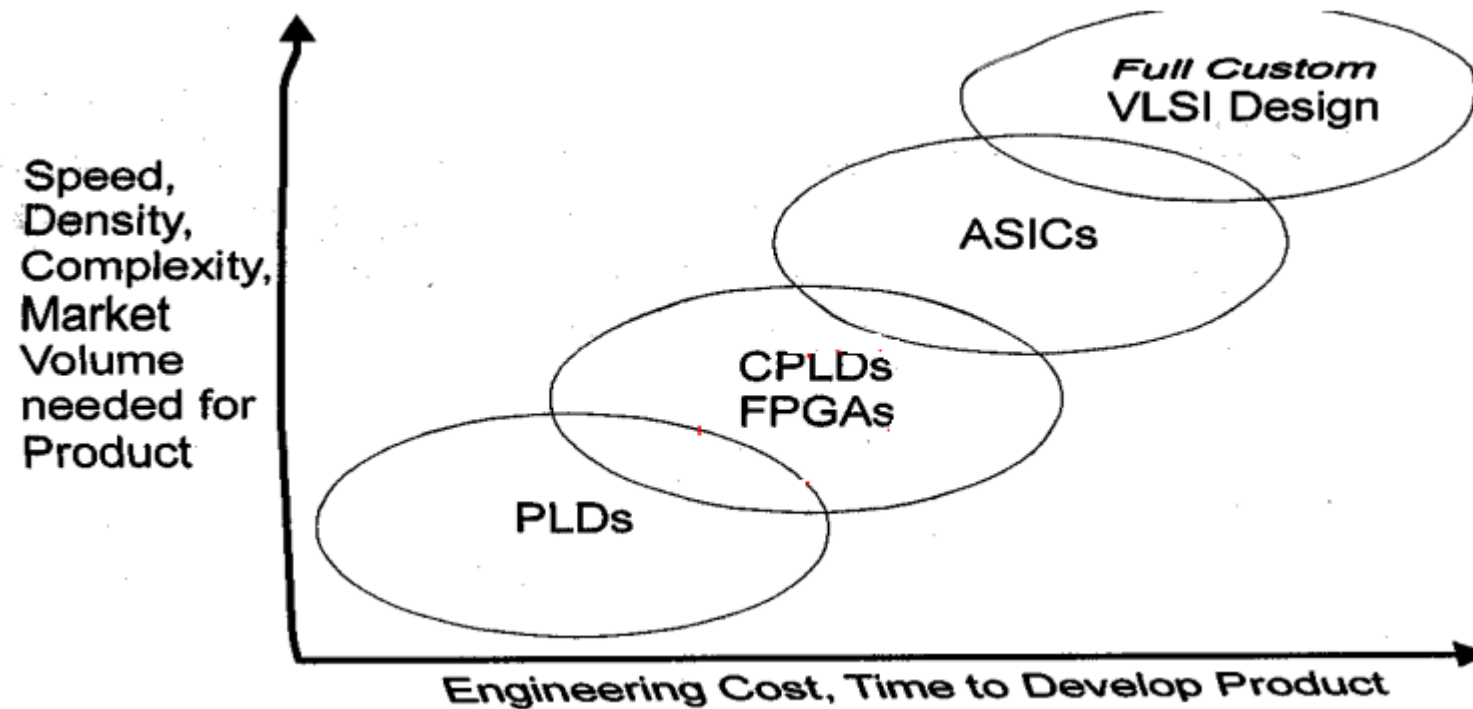
$$Y = A'B + CD + B'D'$$

$$Z = ABC' + A'B'CD' + AC'D' + A'B'C'D$$

$$= W + AC'D' + A'B'C'D$$



DIGITAL LOGIC



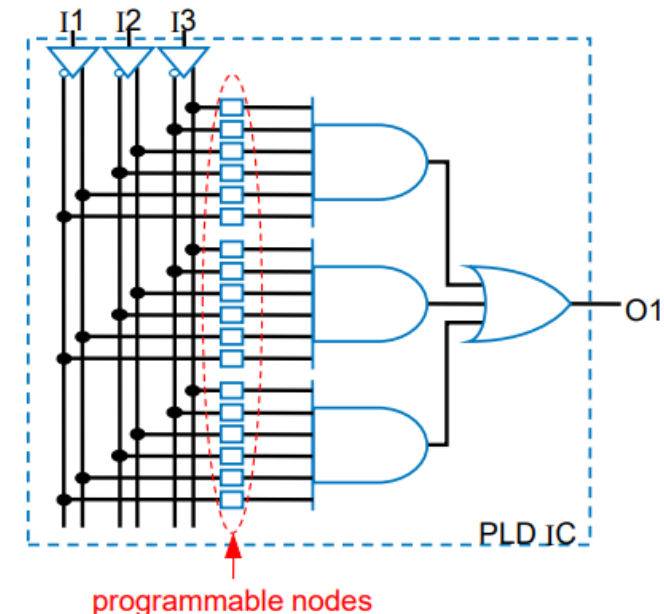
LOGIC DEVICES

- SPLD (Sequential or Simple Programmable Logic Device)
- CPLD (Complex Programmable Logic Device)
- FPGA (Field Programmable Logic Device)

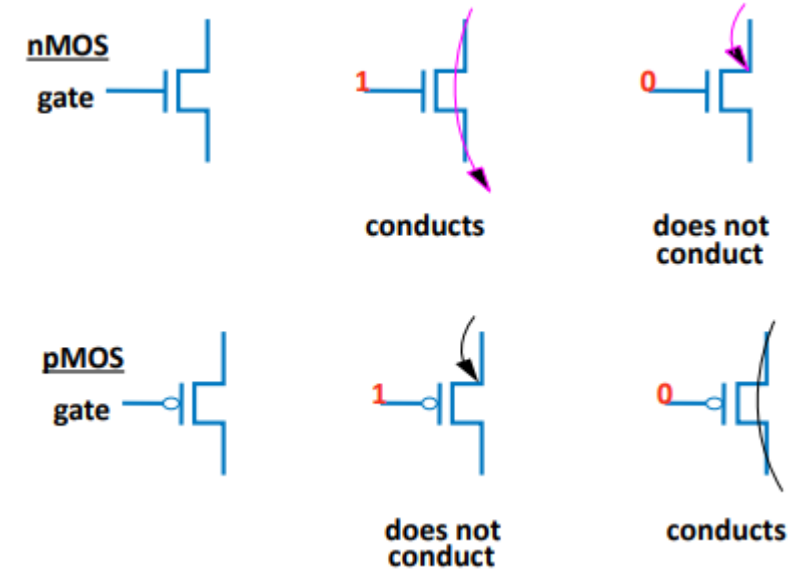
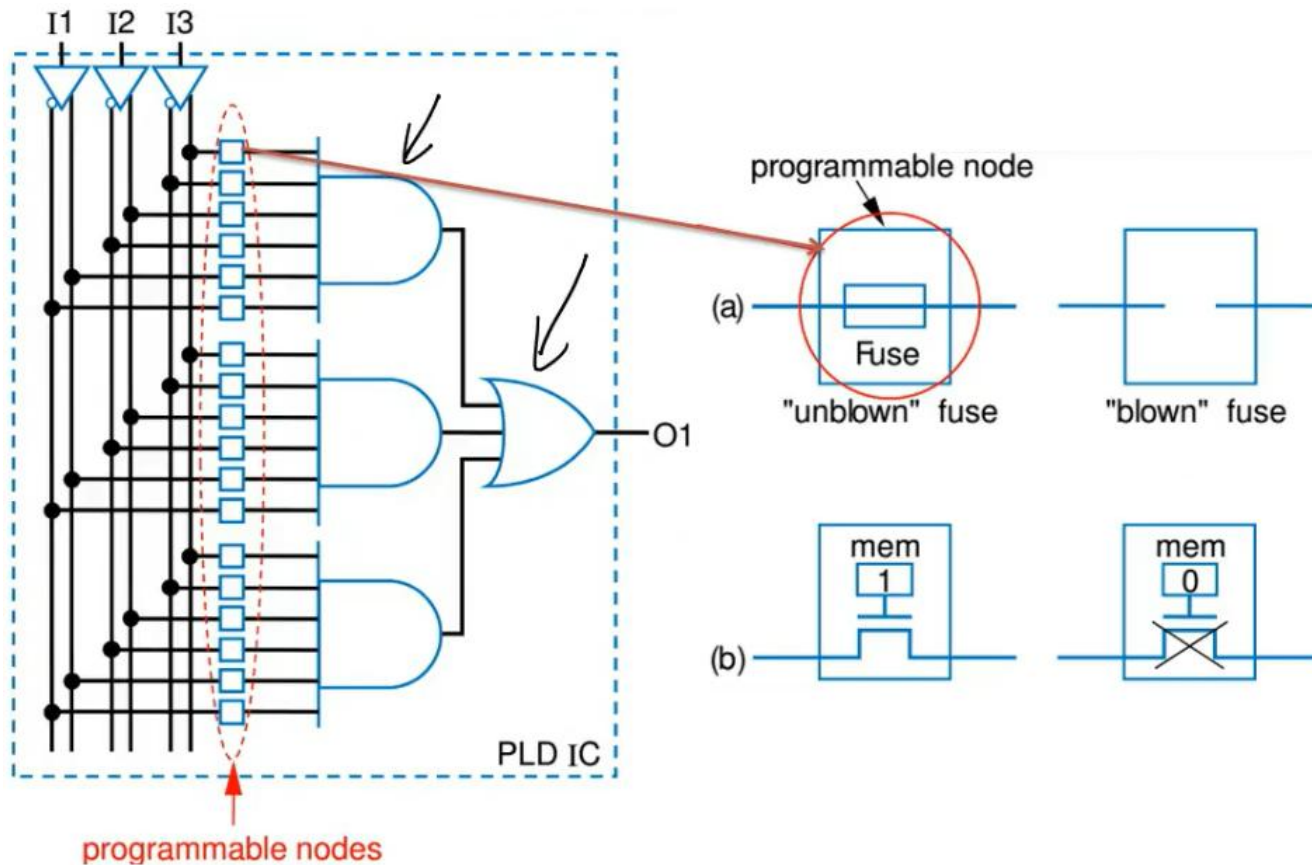


SPLD

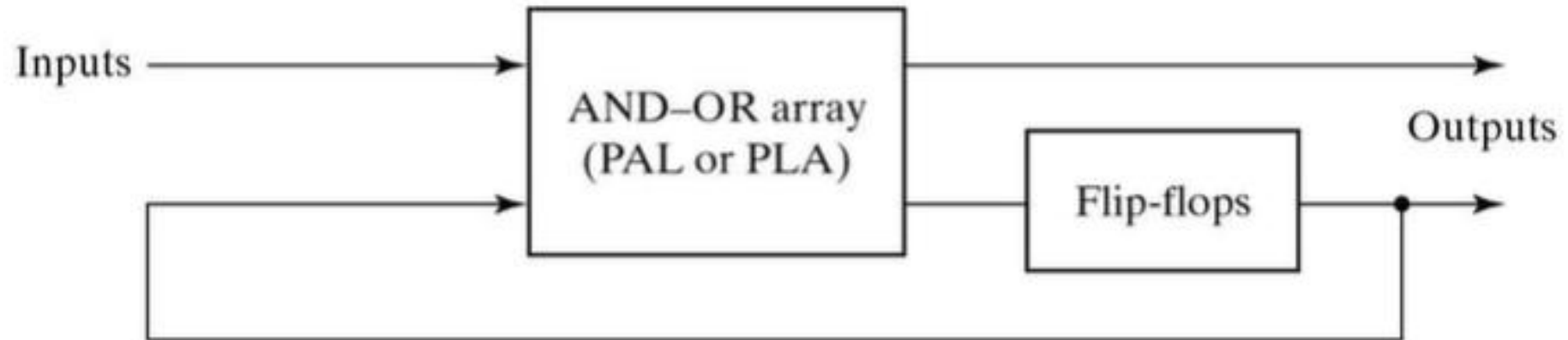
- Developed 1970s (thus, pre-dates FPGAs)
- Prefabricated IC with large AND-OR structure
- Connections can be "programmed" to create custom circuit
 - Circuit shown can implement any 3- input function of up to 3 terms



SPLD

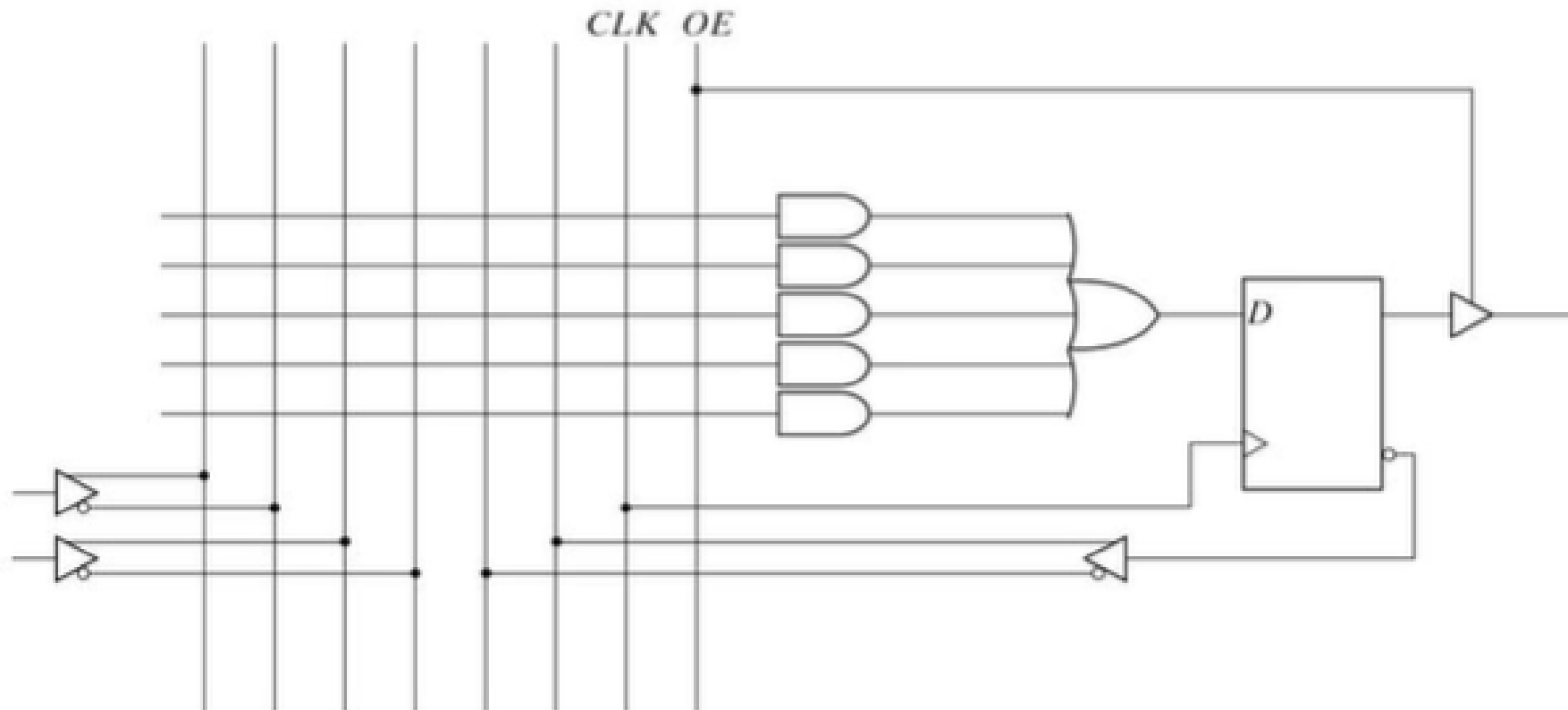


SPLD

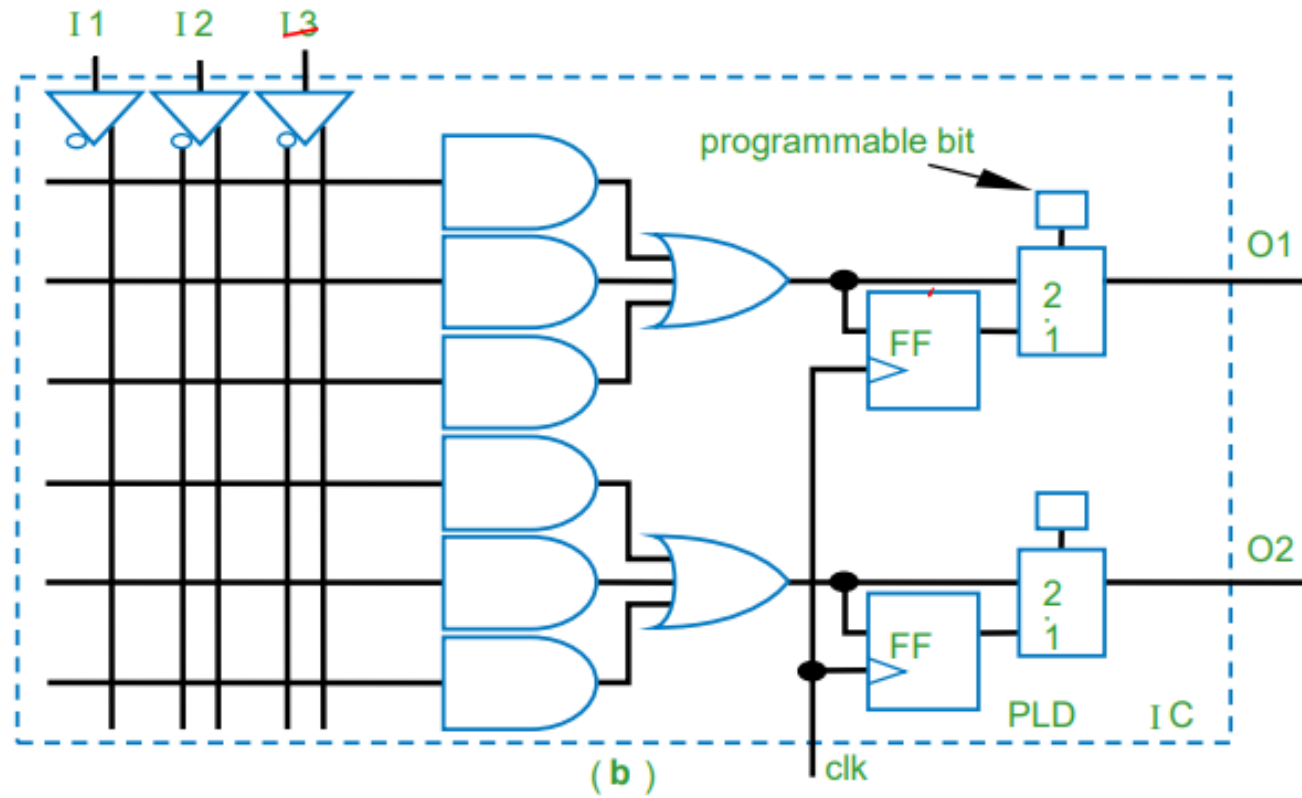


MACROCELL

- Each section of SPLD is called a macrocell.



MACROCELL



Macrocell

